

DDA5001-L03 Machine Learning

The Chinese University of Hong Kong, Shenzhen
Fall 2026

Course Information

Instructor	Junchi Yang
Email	yangjunchi@cuhk.edu.cn
Office	Dao Yuan 321a
Office Hours	Wednesday, 2:00–3:00 p.m.
Lecture Time	Wednesday and Friday, 10:30–11:50 a.m.
Lecture Location	Zhi Xin Building 111

Teaching Assistants

Name	Email
Mingze Kong	225040539@link.cuhk.edu.cn
Jiahang Sun	225045021@link.cuhk.edu.cn
Zhiwei Shang	225040503@link.cuhk.edu.cn
Pingchen Lu	2720801657@qq.com
Zikun Qu	225040504@link.cuhk.edu.cn
Jiarun Shen	225045001@link.cuhk.edu.cn
Mengqi Li	223040115@link.cuhk.edu.cn
Yanwen Liu	225040486@link.cuhk.edu.cn

Students are welcome to contact the teaching assistants by email with course-related questions. When necessary, students may also schedule an online or in-person meeting with a TA by appointment for further discussion.

Recommended Reference Books

1. Abu-Mostafa, Y. S., Magdon-Ismail, M., & Lin, H. T. (2012). *Learning from Data*. AMLBook.
2. Bishop, C. M., & Bishop, H. (2024). *Deep Learning: Foundations and Concepts*. Springer.

These books are recommended references rather than required textbooks.

Course Summary

This course covers fundamental topics in supervised and unsupervised learning. The supervised-learning component includes linear regression, least squares, maximum likelihood estimation, train-

ing versus testing, overfitting, cross-validation, logistic regression, gradient-based optimization methods, and regularization. The unsupervised-learning component includes principal component analysis (PCA), its variants, and clustering.

The course also covers advanced topics, including stochastic gradient methods, the Adam family of optimization methods, fundamentals of deep learning, transformers, and language models, with additional emphasis on large language models (LLMs).

Tutorials

The course is tentatively planned to include **four tutorials**. These tutorials **do not take place every week**. The tentative schedule in the original course plan is:

- **Tutorial 1:** 2nd teaching week (September 15). Topics include installing and configuring OpenCode and Codex, together with an introduction to code generation, debugging, repository navigation, and agentic task execution.
- **Tutorials 2–4:** 9th, 11th, and 13th teaching weeks (November 10, November 24, and December 8). These tutorials focus on the course project and its large-language-model components.

There will be no separate tutorials devoted to Python, individual learning algorithms, or homework walkthroughs. Students are expected to use appropriate coding tools to resolve routine technical and assignment-related questions independently, subject to the course academic-honesty policy.

Tutorial time and venue: 6:30–7:30 p.m., online via Zoom. The Zoom link will be announced on Blackboard.

Homework

There will be approximately **three homework assignments**. Homework must be submitted through Blackboard by the stated deadline. **Late submissions will not be accepted** unless an excused absence has been approved in advance under the policy below.

Course Project

The course includes a project on large language models (LLMs).

Project Structure

- The project consists of **three consecutive parts**.
- Parts I and II each have approximately **two weeks** for completion. Part III has approximately **three weeks**, including the final review and exam period. The full project therefore spans about seven weeks.
- The project is tentatively scheduled to be released around the 8th teaching week (November 8).
- Students may work **individually or in groups of two**. **Part I must be completed individually**. Parts II and III may be completed with one partner; students who choose to work individually may also complete these parts on their own.

- For group work, each member must clearly state their responsibilities and contributions.

Submission Materials

Assessment will be based on the following materials:

- Part I: PDF report, up to 1 page.
- Part II: PDF report, up to 2 pages.
- Part III: PDF report, up to 2 pages.
- References pages are not counted toward the page limits.
- Code submission is required.

Project Grading

The project is worth 100 points in total:

Part	Points
Part I	30
Part II	40
Part III	30

Project Tutorials

Tutorials 2–4 are intended to introduce the large-language-model components of the project and provide guidance for completing the project.

Project Computing Resources

The School will provide on-site computing resources with GPU-equipped workstations for the course project. Students or groups without suitable computing resources will need to use the designated on-site facilities. Detailed instructions for accessing and using these resources will be announced later.

The provided computing resources may be used only for course-project activities. Unauthorized use is strictly prohibited, and unusual usage will be recorded.

Assessment and Grade Distribution

Component	Weight
Homework	18%
Course Project	30%
Final Exam	52%

Final exam: 3:00–5:00 p.m., Saturday, December 19, 2026.

Course Policies

Exams

If a student misses an exam without prior permission, a score of **0** will be assigned for that exam, and this information will be reported to the Registry.

Students who anticipate a conflict with an exam time should contact the instructor or the TAs as early as possible.

Assignment Submission and Excused Absences

- Students seeking an excused absence for an assignment must contact the instructor before the assignment deadline and provide appropriate supporting documentation.
- Excused absences will be granted only with valid documentation for medical reasons or approved University-related activities.
- An assignment that is not submitted and does not have an approved excused absence will receive a score of **0**.

Grading Inquiries

- Questions or discrepancies regarding homework grading should be raised with the TA within one week after the corresponding score is released.
- Grading inquiries submitted after the one-week period will not be considered.

Academic Honesty

- Academic integrity is required in all course work.
- Academic dishonesty includes, but is not limited to, copying homework solutions, cheating on exams, using unauthorized materials, and submitting work that is not one's own.
- Students may discuss homework problems with classmates, but submitted solutions must be written independently unless an assignment explicitly permits collaboration.
- CUHK-Shenzhen adopts a policy of **zero tolerance** toward academic dishonesty.
- For University regulations and additional information, see the academic-honesty page: <https://registry.cuhk.edu.cn/en/page/30>.